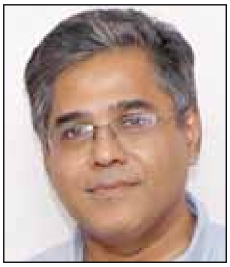
Experts



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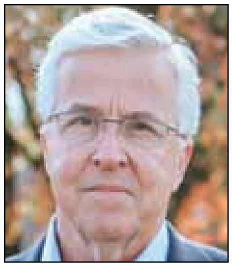
L. Venkata Subramaniam, Senior Manager - AI Science and IBM Quantum Ambassador, IBM Research India

Rahul Mahajan, CTO, Nagarro

Prof. Rainer Dumke, Principal Investigator at the Centre for Quantum Technologies, Professor at Nanyang Technological University, Singapore, and Lead for hardware at Singapore's National Quantum Computing Hub



Robert Liscouski, CEO, Quantum Computing Inc. (QCI)

wealth management, logistic companies attempting to reduce costs and improve efficiencies, and pharma companies that need to tackle the quantum chemistry of their new molecules. We also expect telecom and data security companies to be watchful of any breakthroughs in cryptography, and hence invest in quantum communications and post quantum cryptography," says Prof. Prabhakar.

"IIT Madras has been an early mover in quantum science and technology. We had India's first quantum key distribution demonstration as early as 2008. Today, we have about 20 faculty and their students and staff looking at different aspects of quantum science and technology. This has developed into a healthy interdisciplinary ecosystem, and is formally known as the Centre for Quantum Information, Communication and Computing. We continue to drive both basic and translational research in collaboration with start-ups and our partners in an Industry-Academia Consortium model, allowing members to leverage common knowledge while pushing specific applications of interest to them," he adds.

A good quantum of hope

"Hardware with 100+ logical qubits is now available. Hardware with a few 1000s of qubits is not that far, and we may see such machines in the next 2-3 years. On the software part, we will continue to see improvements in the quantum programming platforms," says Mahajan.

"With the current noisy systems, we are only able to solve some artificial toy models without practical relevance. However, I am sure that in the next generation of systems we will be able to really realise quantum processors that make an impact and can outperform their classical counterparts on real world problems," remarks Prof. Dumke.

According to a recent NASSCOM report, India too is gearing up for the quantum era, with about 10-15 government agencies, 20-30 service providers, 15-20 startups and 40-50 academic institutions active in this domain. It says that the industry would add about $310 billion to the Indian economy by 2030 and sectors such as manufacturing, high-tech, banking, and defence will likely lead the charge of adopting quantum technologies for critical and large-scale use cases.

Achyuta Ghosh, Research Head at NASSCOM, said in an Economic Times interview, that companies like Tata Consultancy Services, HCL Technologies, Infosys, Tech Mahindra, Zensar, Mphasis, and Coforge were creating use cases for quantum technologies and proof-of-concept for clients. There are also some interesting startups in this space such as QNu Labs, Taqbit Labs, QRDLab, and QpiAI Tech.

There are around a 100 quantum projects initiated in India, of which 92% are funded by the government. In August this year, the Ministry of Defence announced that the Indian Army had started the process of procurement of Quantum Key Distribution (QKD) technology developed by a Bengaluru based cybersecurity company by issuing a commercial request for proposal.

In a global survey of 750 companies conducted by KPMG early this year, a quarter of the business decision-makers said they already had quantum computing projects in place, one-third had either an internal team or external advisers looking into how they can use the technology, and 31% of organisations said they were discussing how they will leverage the technology in the future. KPMG reported that only 9% of the respondents said they were not thinking at all about how they can take advantage of quantum. We will leave you thinking about that! EFY